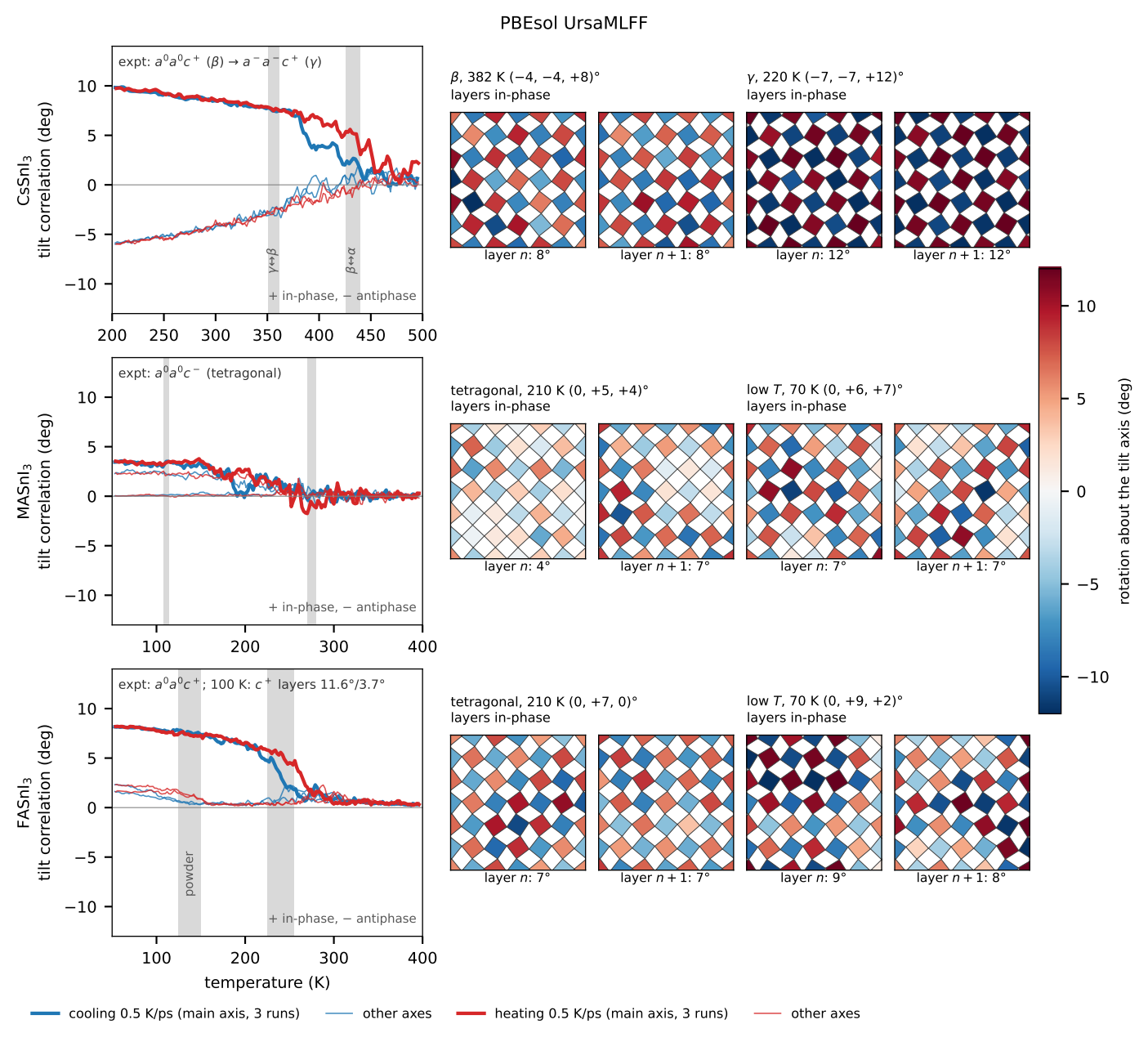

PBEsol UrsaMLFF
expt: $a^0a^0c^+$ (β) → $a^-a^-c^+$ (γ)
$CsSnI_3$ tilt correlation (deg)
$\gamma \leftrightarrow \beta$
$\beta \leftrightarrow \alpha$
+ in-phase, − antiphase
β, 382 K (−4, −4, +8)°
layers in-phase
γ, 220 K (−7, −7, +12)°
layers in-phase
layer n: 8°
layer $n+1$: 8°
layer n: 12°
layer $n+1$: 12°
expt: $a^0a^0c^-$ (tetragonal)
$MASnI_3$ tilt correlation (deg)
tetragonal, 210 K (0, +5, +4)°
layers in-phase
low T, 70 K (0, +6, +7)°
layers in-phase
layer n: 4°
layer $n+1$: 7°
layer n: 7°
layer $n+1$: 7°
expt: $a^0a^0c^+$; 100 K: c^+ layers 11.6°/3.7°
$FASnI_3$ tilt correlation (deg)
powder
tetragonal, 210 K (0, +7, 0)°
layers in-phase
low T, 70 K (0, +9, +2)°
layers in-phase
layer n: 7°
layer $n+1$: 7°
layer n: 9°
layer $n+1$: 8°
rotation about the tilt axis (deg)
temperature (K)
cooling 0.5 K/ps (main axis, 3 runs)
other axes
heating 0.5 K/ps (main axis, 3 runs)
other axes